

SF2943 - Time Series Analysis Project

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Time Series Analysis of Historic Temperature Data

Introduction This report presents the findings from the analysis of a temperature time series dataset, focusing on historical temperature data ranging from January 1756 until December 2023, with an additional dataset to validate our forecasting. The primary objective was to apply time series analysis techniques to forecast future temperature values and assess the robustness and reliability of these predictions. All the data exploration and analysis was conducted in the R programming language.

Data Description The dataset comprises monthly mean temperature readings recorded at Stockholm, Odenplan, retrieved from SMHI's public database. The validity of the historical data is verified by SMHI, and can be found at the SMHI website.

Exploratory Data Analysis and Preparation We begin by eliminating irrelevant columns and transforming data into a time series object, using the built-in `ts` method. Furthermore, the data was already nicely preprocessed, hence no missing values was detected. Below we visualise the whole time series data from January 1756 to December 2023 as well as the ACF and PACF showing line (1).

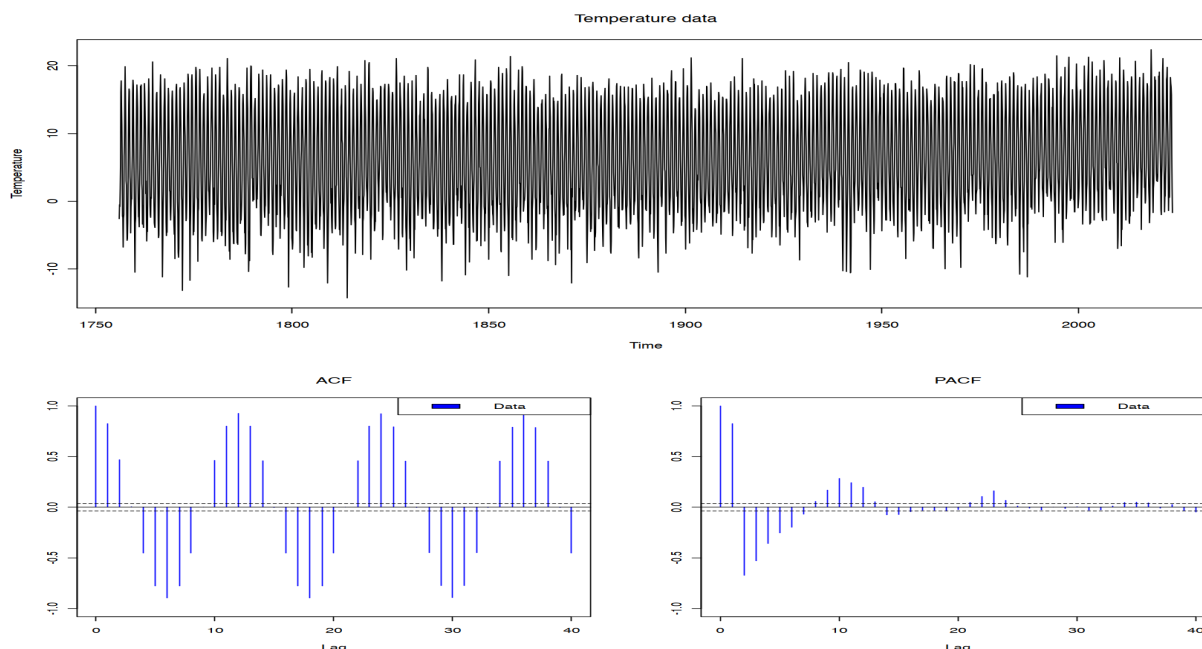


Figure 1: Original time series data of monthly average temperature in Celsius in the Stockholm region. ACF and PACF plotted for the entire series shows clear yearly seasonality traits.

Continuing by splitting the data into train and test sets with a cut-off between December 2009 and January 2010, we aim to evaluate our forecast on the last 14 years of the data. Possibly, a slight positive trend can be detected in figure (1), which is likely given the on going climate change, and can likely be approximated as a linear function. Following this, we used `trend`, `season` and `Resid` from the `itsmr` library to decompose the time series into three components: trend, seasonality, and the remainders. The seasonality was set to capture the recurrent behaviour every 12th month, capturing the yearly seasonality indicated by the ACF plot, while the trend was represented by a linear model, as mentioned. The extracted seasonality and trend is illustrated below in figure (2), confirming the time series is indeed non-stationary.

Moreover, an Augmented Dickey-Fuller test, using the `tsseries::adf.test` method, was made on the

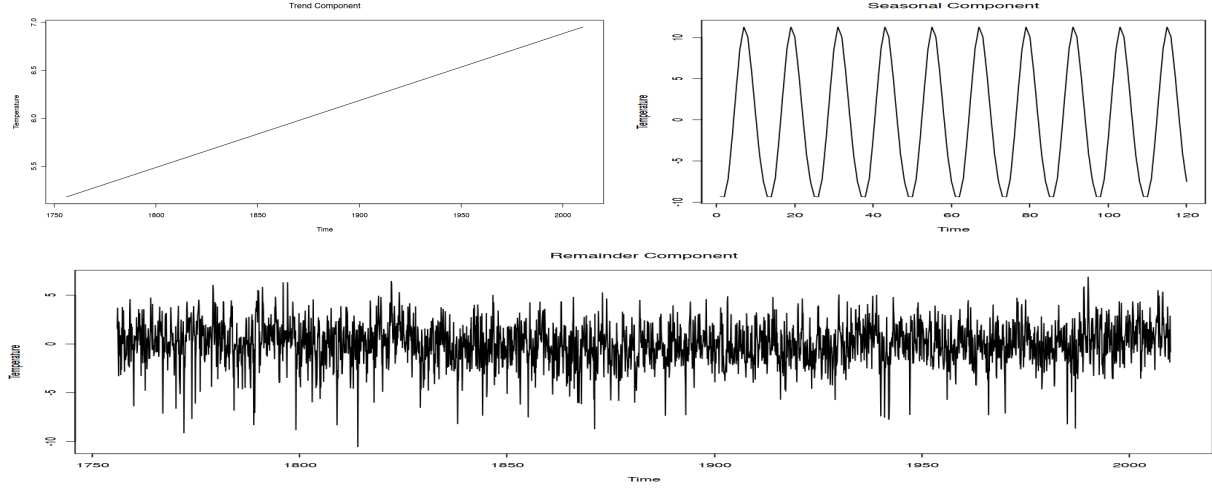


Figure 2: Plots of the detected linear trend and seasonality extracted from the training time series as well as the remainder, which shows white noise behaviour.

remainders, showing enough evidence to reject the null hypotheses of any zeros in the complex unit disc, hence nothing indicates non-stationarity and we may, thus, assume the remainder is stationary, which is also affirmed visually. The ACF, which can be seen decaying exponentially in figure (3) below, provides further evidence that the non-stationary parts have successfully been extracted. It does show, however, traits of some fluctuation, although within or around the confidence interval, which might indicate some complex pattern in the series not captured in the process. Furthermore, the QQ-plot also indicates a quite good preprocessing but does also, however, show a slight heavy tail further indicating the plausible complex pattern not captured.

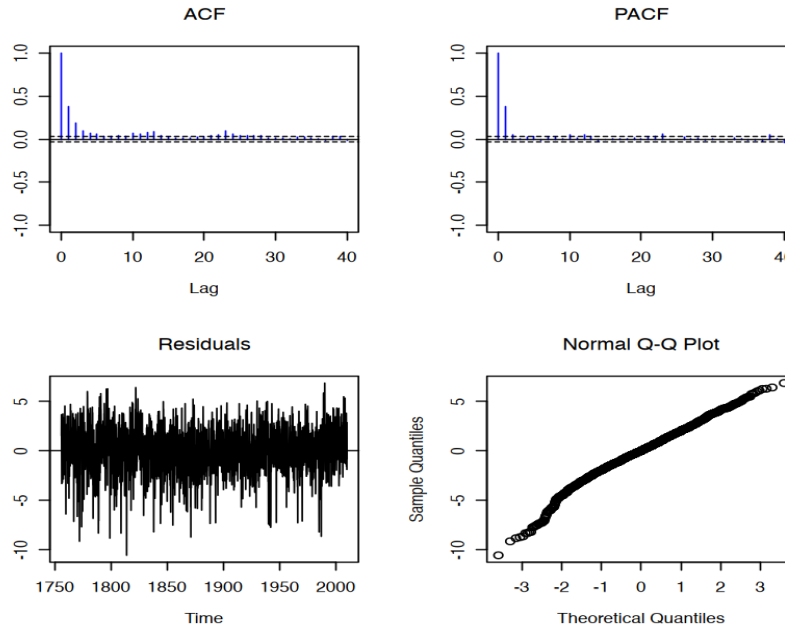


Figure 3: Residual analysis on the training data, showing a decent extraction of the non-stationary properties.

Model Selection and Fitting To determine the $ARMA(p, q)$, we inspect the ACF and PACF in figure (3). Since the ACF decays quickly and shows no clear dependencies above a lag of 2, and the PACF cuts off sharply after the first or second lag, we can infer that $AR(1)$, $AR(2)$ and possibly an $MA(1)$, hence also a possibility of an $ARMA(p, 1)$, for $p = 1, 2$. We do, however, make a grid search comparison for $p \in \{1, \dots, 5\}$ and $q \in \{0, \dots, 4\}$, to inspect if there are alternative models within these bounds providing a better result.

Furthermore, to evaluate and compare the above models, we fit them to the data using the MLE-based method `arma` from the `itsmr`-package. To compare the models we use the Akaike Information Criterion (AIC), which is visualised in the heatmap in figure (4) for $p \in \{1, \dots, 5\}$ and $q \in \{0, \dots, 4\}$. As seen in the heatmap below, $ARMA(2, 1)$ and $ARMA(5, 3)$ can be of particular interest given the simplicity of $ARMA(2, 1)$ and the best AICc value of $ARMA(5, 3)$. However, given that the difference in AICc is not significant, we choose, in line with Occam’s razor, the simpler model $ARMA(2, 1)$ to hopefully decrease the risk of overfitting. We do, however, keep the $ARMA(5, 3)$ to see if the chosen model did generalise better than the more complex one, by comparing prediction error metrics. Furthermore, to evaluate the $ARMA(2, 1)$ model, we investigate the residuals of the fitted model, also visualised in figure (4).

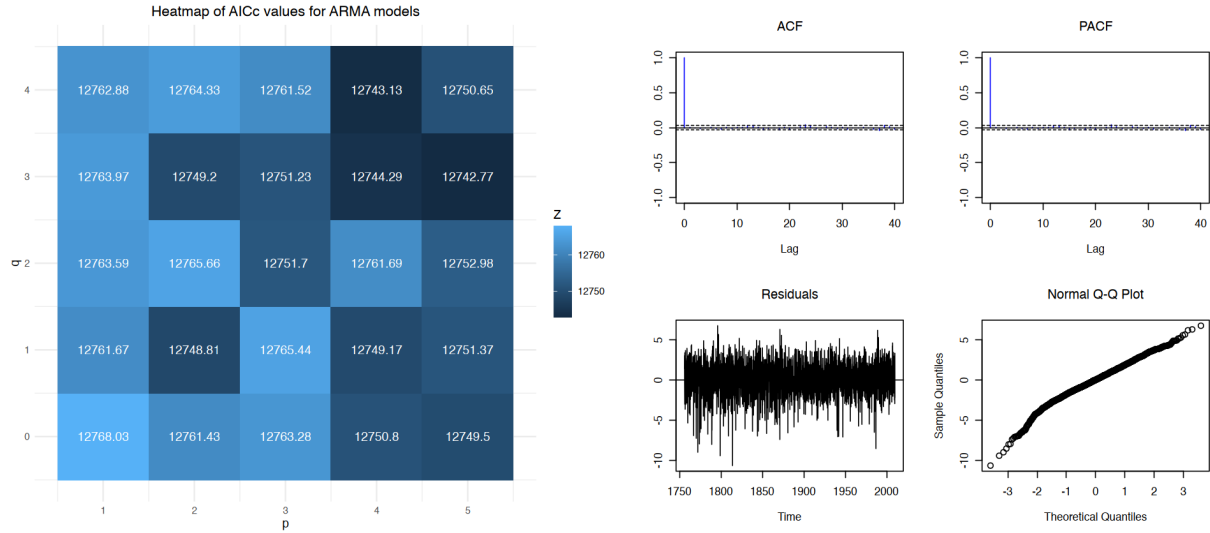


Figure 4: To the left is the heatmap of the AICc of $ARMA(p, q)$ for $p = 1, \dots, 5$ and $q = 0, \dots, 4$. To the right, a residual analysis plot for the fitted $ARMA(2, 1)$ model, showing ACF, PACF, residuals and the QQ-plot.

We can observe that the residuals are reasonably normal with a slight heavy tail, and that they are linearly uncorrelated in between different lags, according to the ACF and PACF plots. Moreover, the parameters of the fitted $ARMA(2, 1)$ model is

$$X_t = 1.315X_{t-1} - 0.3328X_{t-2} + Z_t - 0.9509Z_{t-1}, \quad \text{where } Z_t \sim WN(0, 3.827). \quad (1)$$

Forecasting As a fitted model have been selected, we can now use it to make forecasts on the test set, which we then can evaluate using error metrics such as the MAE and RMSE. The 14 year forecast on the test set of future monthly mean temperature, using the $ARMA(2, 1)$ model is visualised in figure (5) together with the true values and the confidence interval of the forecast.

Overall, the fitted model does capture most of the behaviour of the data, and most of it lies within the confidence interval, except some extreme points, mainly during summer and winter. Although, considering the ongoing climate change, the model could possibly benefit from a more careful extraction of seasonalities and trends. In particular, incorporating a non-linear trend, which might capture the rate of increase in CO_2 -equivalents could possibly provide a model more reflective of the real world.

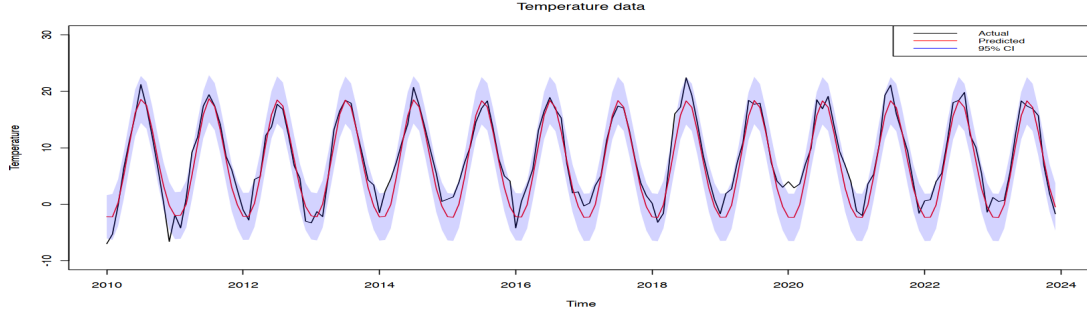


Figure 5: Forecast of the monthly mean temperature for the years 2010 through 2023 using a fitted $ARMA(2, 1)$ model together with its corresponding confidence interval.

Moreover, the chosen model's ability to forecast future temperature can be evaluated by quantifying the prediction errors via the error metrics Mean Absolute Error (MAE) and the Root Mean Square Error (RMSE). These are shown in the table below for $ARMA(5, 3)$ and $ARMA(2, 1)$ model to provide a comparative result of the chosen model.

$$\begin{aligned} MAE_{(2,1)} &= 1.7978 & RMSE_{(2,1)} &= 2.2662 \\ MAE_{(5,3)} &= 1.7269 & RMSE_{(5,3)} &= 2.1841 \end{aligned}$$

The resulting measures of error show that the $ARMA(5, 3)$ model is slightly more accurate than the $ARMA(2, 1)$ in terms of these errors. However, the difference can be considered not very significant, and therefore, the choice of $ARMA(2, 1)$ can be considered legitimate.

Challenges, Limitations and Possible Enhancements The main challenge encountered was to properly extract seasonalities and trends, making the temporal data stationary. Finally, it might be beneficial to explore more complex models like ARIMA and SARIMA to capture more sudden and non-cyclical changes in the temperature data, or possibly state space models such as Gaussian Processes and incorporate more data in such as rate of change in CO_2 -equivalents.

Time Series Analysis of Financial Data

Introduction This report aims to forecast the share price of the SEBInvest AKL Danske Aktier fund using an ARMA model, focusing primarily on the closing price for analysis. The approach involves the removal of trends and seasonal components attempting to make the time series stationary as well as differencing using the 'diff' function in R, attempting to achieve stationarity.

Data Description and Preparation The utilised data consists of the highest price, lowest price, and closing price of the stock recorded on open market days. In this report, the focus is on the closing price, since we regard it as the most relevant objective for the purpose of this analysis. We do, however, note a discrepancy in the length of fiscal years prior to 2004 compared to later years as can be seen in table (1) where financial years prior to 2004 were notably shorter. We hypothesised that one seasonal component might be influenced by the length of a financial year, typically spanning 252 days. Therefore, to mitigate potential interference in our analysis, we made the decision to exclude data from years before 2004.

Fiscal year	2000	2001	2002	2003	2004	2005	...	2022	2023	2024
Number of data points	29	119	91	177	252	252	...	252	250	69

Table 1: The number of data points to the corresponding financial year.

Below, in figure (6), the time series is visualised in its entirety using `ggplot2`, for every objective, Low price, High price and Closing price.

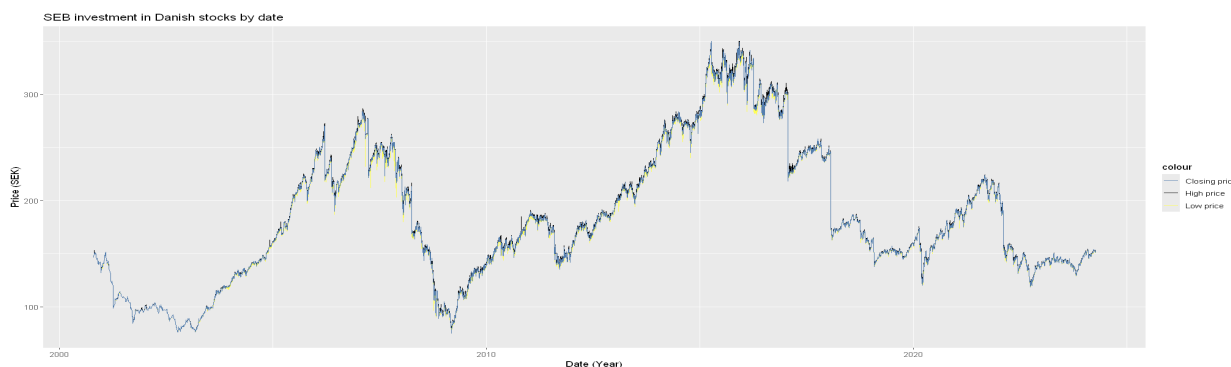


Figure 6: Low, High and Closing price for the SEBInvest AKL Danske Aktier ranging 2000 to 2024.

All together, the data used in this report is the closing price from financial years posterior to 2003. Furthermore, the data was divided into a training and a test set, where the training consists of the data up until 2024 while the test set contains the 69 points of 2024.

Exploratory Data Analysis Obviously, the time series is non-stationary, hence preprocessing of the data has to be made before fitting an *ARMA*-process to it. Using the method `stl` from the `forecast` library in R, we decompose the time series into seasonal, trend and remainder components. Due to the heteroscedastic behaviour of the remainder, the `BoxCox` function from the `forecast` library was used to perform a Box-Cox transformation on it. The decomposed time series as well as the ACF and PACF for the Box-Cox transformed remainder is visualised in (7) below.

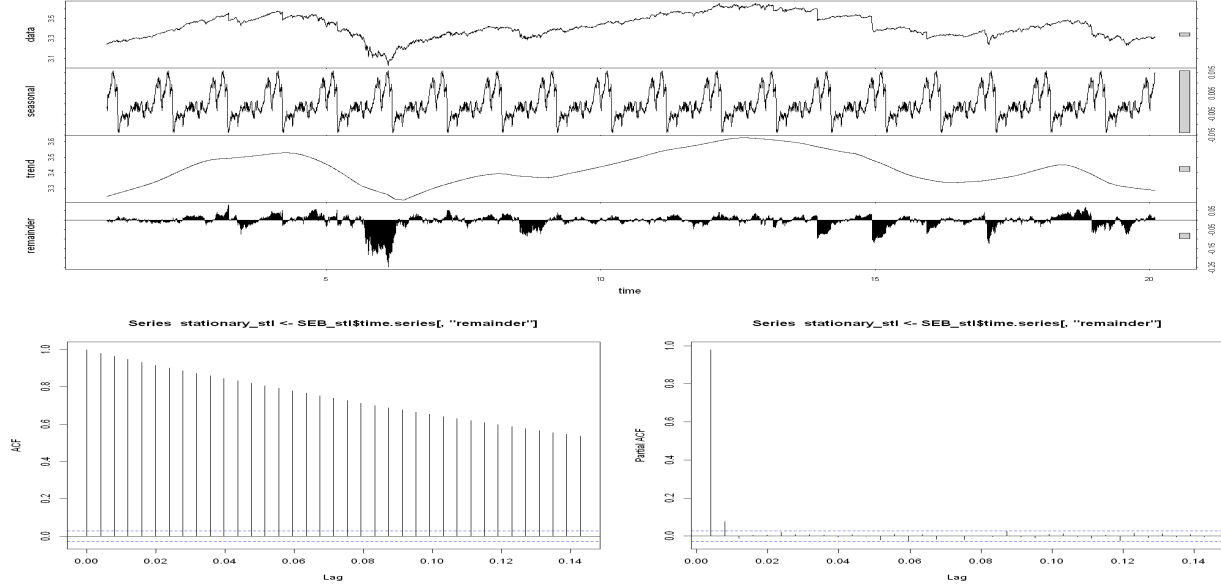


Figure 7: Decomposition of the SEBInvest AKL Danske Aktier time series into seasonality, trend and remainder, which shows a heteroscedastic behaviour. Below this, the ACF and PACF of the remainder, showing a significant autocorrelation.

Moreover, the significant autocorrelation is prominent in figure (7), despite the extracted seasonalities and trends, therefore, we further differencing the remainders with a lag of 1, resulting in the QQ-plot and the ACF and PACF plots in figure (8).

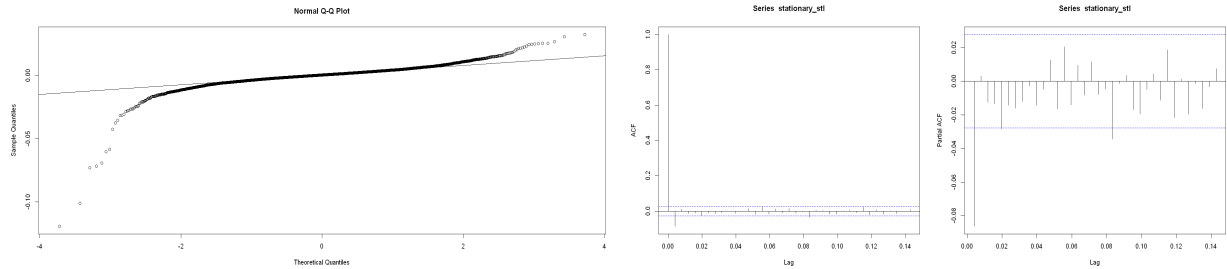


Figure 8: QQ-plot of remainders as well as ACF and PACF plots after differencing with lag 1.

As seen, the remainder shows to be heavy-tailed but mostly normal distribution and the ACF and PACF plots showed essentially no autocorrelation and no significant lags. To check for any clear evidence against stationarity, we run a Dickey-Fuller test, which showed that we could reject the null hypothesis of any unit roots and can, thus, conclude that no evidence for non-stationarity has been found.

Model selection and Fitting In order to select a proper model we performed a grid search over the parameters p and q to determine the number of auto regressive and moving average parts to use. The grid search and model fitting was made using `auto.arima` from the `forecast` package, where the integrating parameter was set to zero. The method compares the models by their corresponding AIC, AICc and BIC values, and return the model with the best overall goodness. This resulted in a $MA(1)$ model. In particular,

the model with its corresponding coefficients is expressed as

$$X_t = Z_t - 0.0855Z_{t-1}, \quad \text{where } Z_t \sim WN(0, 3.627 \cdot 10^{-5}). \quad (2)$$

The residual analysis of this model can be seen in the figure (9) below, showing that it does capture most of the linearly temporal correlations with a normally distributed residual with a slight heavy tale with a few outliers, which was quite expected. This is considered as a model fitted well enough to use for the forecasting.

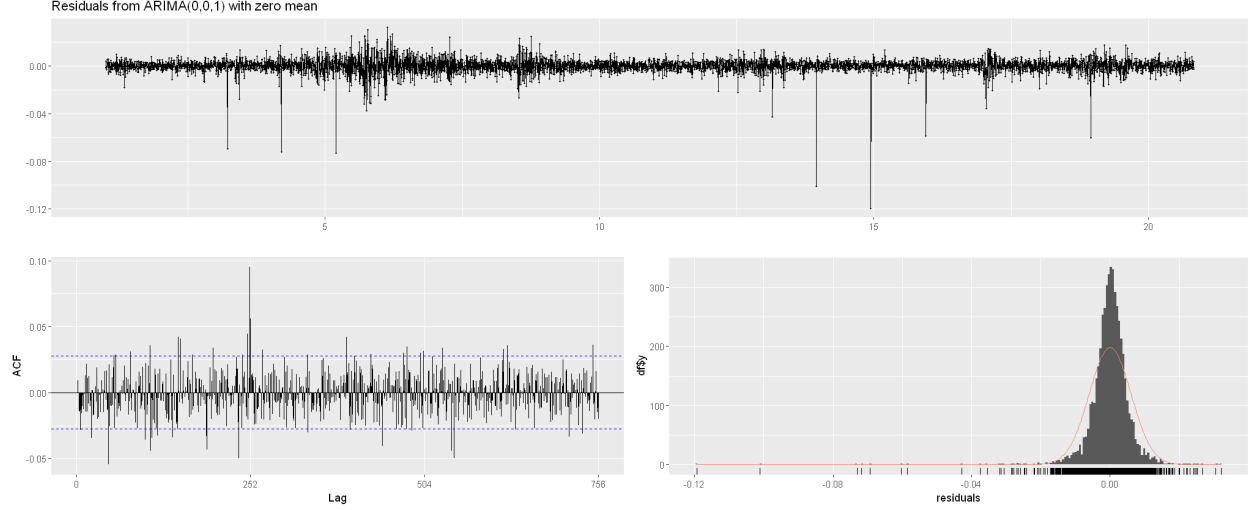


Figure 9: Residuals of the fitted Moving Average ($MA(1)$) model to the closing price data.

Forecasting When making forecasts with the fitted $MA(1)$ model, an inverse Box-Cox transformation is applied to restore the predicted values to their original scale followed by a cumulative sum to reverse the effects of the differencing. Finally, the trend and seasonal components were reintroduced, which resulted in the forecast shown in figure (10). The forecast is plotted together with its corresponding 80% and 95% confidence intervals for a visual representation of the quantified uncertainty of the $MA(1)$ model.

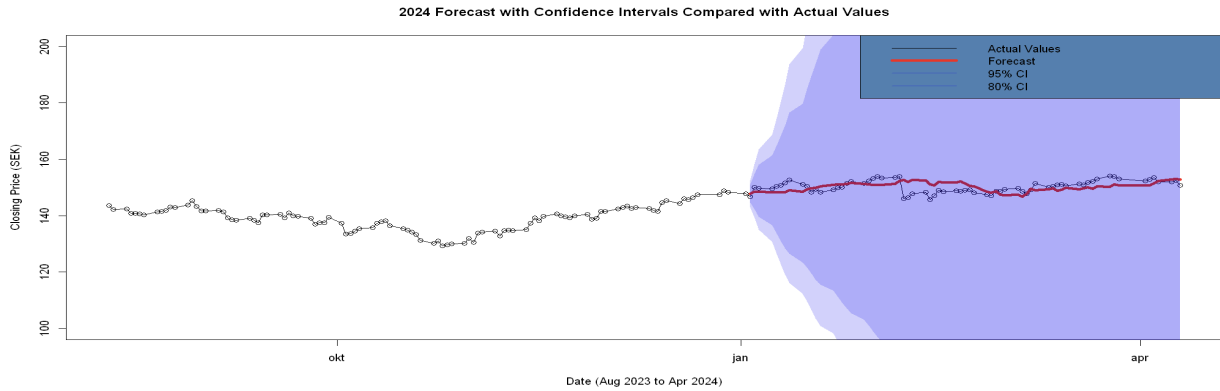


Figure 10: The resulting forecast from January 2024 using the $MA(1)$ model with its corresponding 80% as well as 90% confidence intervals.

Model evaluation From the plotted forecast we observe that the forecast of the closing price follows the test data reasonably well, indicating that an MA(1) model might have been a good fit for the time series. However, given the natural fluctuations of financial data and the relative stable 2024 data, this is a significant simplification and it is likely the model will not capture any sudden changes. Furthermore, this is likely one of the causing factors of the wide confidence intervals.

Moreover, the prediction error of the forecast was calculated with means of root mean square error and mean absolute error for the approximately four and a half months forecast. The computed errors is shown in the table below.

RMSE	2.428501
MAE	2.033453

Table 2: Prediction errors of the MA(1) model

Considering the size of the errors for such a long forecast as 4 months, one might think the model is a good predictor. However, since the model has no auto regressive part it will likely not predict future fluctuations that well which, as mentioned, is likely to be present in financial data.

Challenges and Limitations One major challenge was to remove the seasonality. By simply plotting the time series we could observe that the larger peaks in the data span around 1500 days. However, we could also note many other seasonal traits over shorter periods, which made it difficult to determine a seasonal pattern. Even when plotting the data we could only see a seasonality with around 3 periods, meaning that the season is 1500 days long. We concluded that this seasonality is not worth using in the model, since the period for the season was too big for the data set. The sudden dips in the seasonality aligns well with the dotcom bubble around year 2000, the finance crisis of 2008 and COVID-19, which, hence, likely affected these. In addition, the use of more complicated models such as SARIMA might have been beneficial from a forecasting precision point of view. However, we did not consider those since we specifically wanted to implement an ARMA model. Lastly, we encountered some difficulty in restoring the generated confidence intervals, which might have affected the size of the intervals shown in figure (10).